Janavahini: route-aware crowd alerts that warn the road before the ghat fills

Low-cost counters on every road, gate and junction estimate crowd density, look a minute ahead, and pass warnings back along the pilgrim route to the police who can act, even if the network or the command centre goes down.

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Challenge RTIH × RMC Innovation-Led Solutions for Godavari Pushkaralu 2027 **Date** November 2026

Demo video submitted with this application (PS01_Janavahini.mp4)

Janavahini (జనవాహిని) means "a river of people": the system keeps the river of pilgrims flowing safely to the Godavari. Companion documents: **Technical architecture and workflow** · **Indicative budget and implementation plan** · **Demonstration video**.

SUMMARY

A crush at a bathing ghat builds when more people walk in than can leave. The police who can stop it stand on the approach roads, often a kilometre away, and cannot see the ghat filling. We split each road into short **patches**, each watched by a small node. The node measures how crowded its patch is (line counters where barricades already form lanes, a density reading from existing CCTV or an edge camera on wide roads, fused so counting errors cannot pile up), estimates how full it will be a minute later, and knows which ghats its road leads to. When a ghat is about to cross the safe limit, or already has, the warning travels **backwards along the route**, node to node, until it reaches police who can slow people, hold them, or send them to a ghat with room. Every decision is made on the devices themselves, so alerts keep working when the mobile network or the control room fails; the Integrated Command and Control Centre (ICCC) receives everything through a documented API.

0 s
above the safe limit in a simulated ghat surge, vs about 14 min with today's thresholds

5 of 5
surge runs kept safe with no network at all (today's practice: 0 of 5)

~3 min
earlier warning to the road police (172 s); thresholds never warned the road

100 of 100
site-wide surge scenarios safe with realistic, imperfect sensors; a cut-off command centre managed 6 of 20

All results are from simulation (JuPedSim crowd simulator and a whole-site flow model with modelled sensor errors); see §6. Each section below names the evaluation criterion it answers.

1 · THE PROBLEM · PROBLEM–SOLUTION ALIGNMENT

Danger shows up at the ghat; the people who can prevent it are upstream

On the first morning of the 2015 Godavari Pushkaralu, **14 July 2015**, at least **27 pilgrims died** in a crush at Pushkar Ghat, Rajahmundry; most were elderly women [1–3]. Over the 12 days, about **11 crore** people bathed in the Godavari [4]. For 2027 the state expects up to 10 crore pilgrims, and up to **1 crore in a single day**, almost twice the 2015 peak of 58.2 lakh, across 525 ghats [15].

- **The physics is simple.** A ghat can release only so many people per minute (bathers on the steps, narrow exits). When inflow exceeds that, density rises. Above about **4 persons/m²** movement breaks down; at 5–6 persons/m², shock waves can knock people over [5–7].
- **The timing is the problem.** By the time crowding is visible at the gate, the roads behind it are full and nobody can step back. The police who could stop the inflow are at road heads, junctions and holding areas, out of sight of the ghat.
- **Today's tools are late and local.** Hourly CCTV counts and fixed thresholds alarm at the crowded spot itself, and only through the control room. In our simulations, threshold alarms never reached the road police in time (§6).



Pilgrims at the river front, Rajahmundry, Godavari Pushkaralu 2015. Photo: విశ్వనాధ్.బి.కె., CC BY-SA 4.0, Wikimedia Commons.

2 · OUR SOLUTION · INNOVATION AND ORIGINALITY

Count, look ahead, warn back along the route, act where there is room

1 Measure every patch

Each road is split into patches of about 25 m. At gates and exits, where barricades already form lanes, a **line counter** counts people in and out. On wide roads, a **density reading** comes from an existing CCTV camera or an edge camera wired to the node. A small Kalman filter on the node fuses the two, so counting errors cannot pile up (in − out alone drifts). Only counts leave the device: no faces, no stored video, no IDs.

2 Look a minute ahead

Each counter tracks how fast its patch is filling: **forecast = density now + rate of rise × 60 s**. The rate term is used only once a patch is busy (≥ 1.5 persons/m²), so a normal day raises no false alarms.

3 Warn backwards along the route

A one-time route map records which road feeds which junction and ghat, and in what share. A ghat forecast to cross the limit, or already over it, sends its warning upstream counter to counter, weighted by the share of each road's flow heading into trouble. **Roads that don't feed the problem stay green.**

4 Act where there is buffer space

Police see a three-colour light. Holds happen far upstream where people can wait safely, not at the crowded gate. At junctions, pilgrims are **sent to a neighbouring ghat with room**, which is the advice officials gave on 14 July 2015, but minutes earlier [2].

Green let people through | **Amber** slow the flow, or divert at the junction | **Red** hold here

Levels follow the NDMA crowd guide [5].

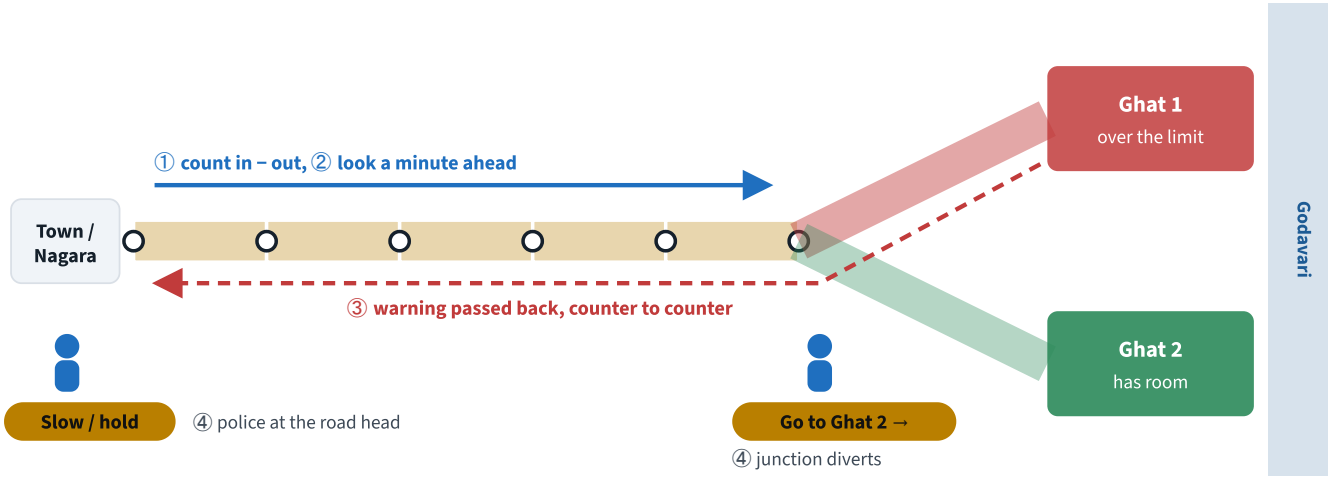


Figure 1. How one alert travels. Each box on the road is one patch; the dots are counters. Ghat 1 is over the limit; its warning reaches the junction, which sends pilgrims to Ghat 2, and the road head, where police slow the inflow. Illustration.

- **Works without the network.** Counters talk to their neighbours over a low-cost radio mesh (ESP-NOW for short hops, LoRa for long ones). Mass gatherings are exactly when mobile networks overload.
- **"Unknown is not safe."** If a board goes silent, its neighbours keep its last warning, listen one hop further, and tell staff which board to check.
- **Makes the ICCC stronger, does not replace it.** When the network is up, each node follows the stronger of the ICCC's command and its own. When the network fails, the mesh carries on.

What is innovative or distinctive

1. **Alerts follow the pilgrim route** (a directed route graph with flow shares), not just the crowded spot, so the warning reaches police who can act about 3 minutes before danger. Earlier designs we tested, averaging and hop-based relays, warned too late or held unrelated roads (§6).
2. **Decided on the devices** over an ESP-NOW/LoRa mesh: works when mobile networks or the ICCC fail, exactly when mass gatherings overload them.
3. **"Unknown is not safe":** a dead node's neighbours keep its warning and flag staff.
4. **Hybrid:** the ICCC can only add caution, never remove it.
5. **Tested against realistic sensors:** biased counters and noisy, missing camera readings, fused on the node. Counters alone fail 15 of 20 surge tests; the fused design is safe in 100 of 100 (§6).
6. **Privacy by design:** counts only. Cameras count on the device; no faces are processed by us and no video is stored.
7. **The firmware logic tested in simulation is the deployed code.**

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3 · TECHNICAL ARCHITECTURE · TECHNICAL FEASIBILITY, SCALABILITY AND CYBERSECURITY

Decisions on the devices; the command centre oversees but is never required

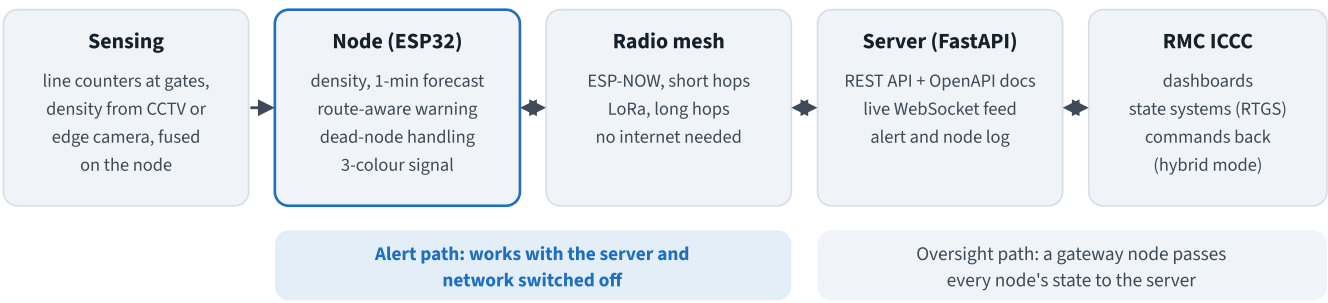


Figure 2. System architecture. Each node takes the stronger of its own decision and the ICCC's command, so the ICCC can only add caution.

Layer	Technology	Status
Sensing	Line counters (a pair of VL53L1X time-of-flight sensors) at gates and exits, where barricades already form lanes; on wide roads, density from existing CCTV or an edge camera wired to the node over LAN/serial, so a network cut does not blind it; a 2-state Kalman filter per node fuses them	Modelled with realistic errors and tested (E12); fusion to be ported to the firmware header
Node	ESP32 firmware (C++, PlatformIO); the alert logic is one header shared with the simulator, so the tested code is the deployed code	Complete; 244 virtual boards tested; not yet flashed
Network	ESP-NOW 25-byte messages once a second (0.7% channel load with 10 boards in range); LoRa SF7 for long hops, sent on change	ESP-NOW done; LoRa stubbed until modules arrive
Server	Python FastAPI; REST with OpenAPI documentation; WebSocket; playback, serial and hardware-in-the-loop modes	Done; end-to-end tests pass
Dashboard	Browser app: zone map, alert log, inflow command, node-down reports with operator clear-down	Done

The full architecture and the step-by-step alert workflow are in the companion document **Technical architecture and workflow**.

4 · HARDWARE, SOFTWARE AND INFRASTRUCTURE · INTEGRATION REQUIREMENTS

Off-the-shelf parts, open-source software, four integration points

Item	Requirement
Hardware per node	ESP32 board with LoRa (SX1276); IP65 three-colour signal tower; IP65 enclosure; mains adapter with backup battery. **Gate and exit nodes:** two VL53L1X time-of-flight sensors, shaded from direct sun. **Wide-road nodes:** a density feed from an existing CCTV camera or an edge camera box, wired to the node.
Quantity	**244 nodes** for the full site (136 gate and exit counter nodes; 108 density nodes at compartments, junctions, roads and Nagaras) plus **10 zone gateways** bridging the mesh to the ICCC over 4G; **11 nodes and a gateway** for the pilot.
Software	Open-source ESP32 firmware, FastAPI backend and web dashboard; runs on one server or VM at the ICCC.
Infrastructure	Power and mounting points at gates, roads and junctions. Counters only where barricades already form lanes (gates, exits); **no road is narrowed** for counting.
Integration	(a) Read crowd counts from existing CCTV/AI analytics, at the node where possible so a network cut does not blind it. (b) Push node states, alerts and node-down events to ICCC dashboards and RTGS via a documented REST/WebSocket API. (c) Accept ICCC inflow commands. (d) Staff messages in Telugu and English.

5 · DATA REQUIREMENTS AND SECURITY MEASURES · CYBERSECURITY

Counts only: nothing personal is collected

- **Data collected:** only counts and densities per patch, node health and alert logs. Cameras count on the device: no images leave it, no video is stored, and no faces, phones or personal data are collected.
- **Data needed from RMC:** model-compartment layout, pedestrian routes, ghat capacities and the CCTV count feed.
- **Built in:** dead-node detection with one staff alert per outage; holds released only by an operator after checking a silent node.

- **Security for the pilot:** authenticated node messages so a spoofed radio cannot raise or clear alerts; TLS on all API traffic; API keys and role-based access for ICCC operators; an audit log of every command; hosting in the RMC or state data centre following CERT-In guidelines.

Tested in a recognised crowd simulator, at whole-site scale, and against the device code

Test	Today's practice	Our system
Ghat surge in the JuPedSim crowd simulator, 5 runs: runs kept below 4 persons/m²	**0 of 5**; peak 6.0, about 14 min above the limit	**5 of 5**; peak 3.3, 0 s above, with 30% radio loss and no network
Same surge, command centre network cut mid-surge	**0 of 5**	**5 of 5** (hybrid; peak 3.1)
Warning reaches the road police	never	**172 s** before the danger point
Whole site: 68 compartments, 20 junctions, 10 Nagaras; 20 random surge scenarios	6 of 20 safe (a cut-off command centre: also 6 of 20)	**20 of 20**
Whole site: 20 blocked-exit scenarios	12 of 20 safe	**20 of 20**; peak 2.3 vs 2.8 for a command centre, with 4× fewer holds because it diverts
Realistic sensing, whole site, 100 surge + 100 blocked-exit draws: biased counters at gates and exits, noisy camera readings on roads, fused by a Kalman filter	counters alone (in − out): **5 of 20** surges safe; the error grows to 5 persons/m²	**100 of 100** safe in both (95% lower bound ≈ 96%); same worst peak (3.57) as perfect sensing
How good a density reading must be (20 draws per setting)	–	safe with a reading every ≤ 2 min and ≤ 20% error; also with half the readings lost or a 20% camera undercount; 30% error fails 2 of 20
A dead board at the gate	first design: about 13 min above the limit	final firmware: **0 s**; also safe with 5% of all boards dead at random, and with realistic sensing
A camera fails while its board runs on	–	safe in every case tested (compartment, gate, junction, road)
A normal day	–	**0** false alerts
Device code vs design model: 244 virtual ESP32 boards, 30% message loss	–	**100%** identical decisions over 1.8 million board-seconds
One-equation model of a filling ghat, not fitted to the test runs	–	danger predicted at **670 s**; simulator measured 672 s

The simulator is JuPedSim (Forschungszentrum Jülich, open source) [8]. Our model passes the RiMEA verification tests 1, 4 and 12 [9] and is calibrated to Weidmann's walking-speed data [10]; every compartment result is repeated over 5 random seeds. Sensor errors (counter bias and noise, camera noise, bias and dropouts) are modelled, not yet measured. The full method, every experiment (including what failed) and the reproduction steps are in our technical report, available on request.

Expected impact during Godavari Pushkaralu 2027

For 2027's up to **1 crore pilgrims a day**, this means:

- **Early holds upstream**, where people can wait safely, instead of at the crowded gate;
- **Diversion to ghats with room** instead of closures, with **4× fewer holds**;
- **Zero false alerts** on a normal day;

- A **live density picture** for the ICCC.

7 · PROTOTYPE · IMPLEMENTATION READINESS

Working software, device firmware and a live demo

Part	Status
ESP32 firmware (counter node, gateway, band tag)	Complete; compiles; uses 22% of RAM. Runs the route alerts and dead-board handling. Tested as 244 virtual boards; **not yet flashed and field-tested**.
Command-centre backend	FastAPI service with a documented REST API (OpenAPI at `/docs`) and a live WebSocket feed; end-to-end tests pass.
Dashboard	Live map of every zone coloured by density and alert, alert log with times, inflow command, dead-node reports.
Demo (24 Nov)	A simulated surge drives real boards. We switch off the server and the road node still turns amber over the mesh; we unplug a node and the road keeps holding, while the dashboard shows "NODE DOWN – send staff".

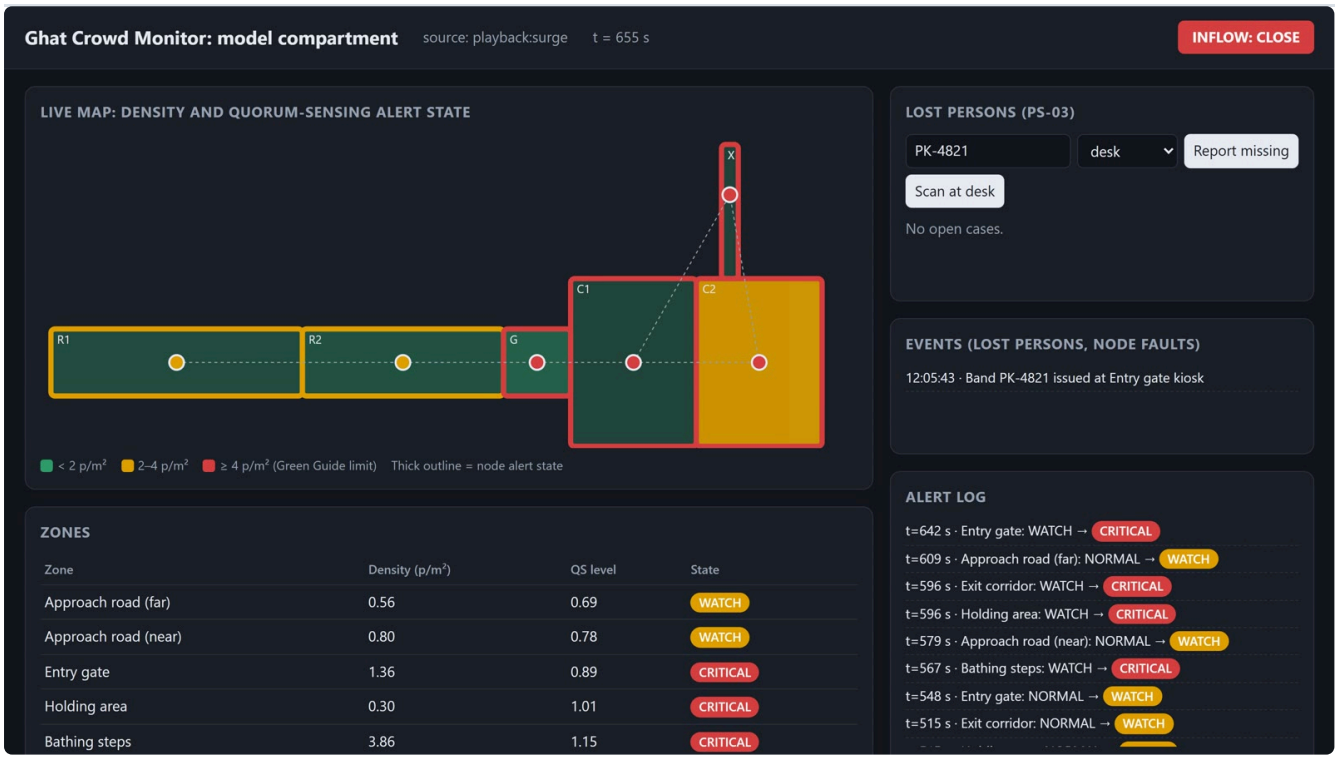


Figure 3. Our prototype dashboard replaying a simulated surge. The warning has reached the approach road (amber) before the bathing steps pass the limit; the alert log records each change with its time, and the inflow command to the road is CLOSE.

8 · SCALABILITY AND SUSTAINABILITY · FIT WITH PUSHKARALU REQUIREMENTS

Add a ghat by adding nodes and edges; reuse every node after the event

- **Scales without retraining.** Adding a ghat means adding nodes and edges to the route map. Tested on a route graph of 68 compartments, 20 junctions and 10 Nagaras (244 virtual boards, 30% radio loss); radio load is 0.7% with 10 boards in range.
- **No central bottleneck.** Decisions are local, so cost grows linearly with the number of nodes.
- **Sustainable.** Hardware is low-cost (₹5,100–8,900 per gate or exit node), off-the-shelf and repairable locally; software is open-source with no licence fees.
- **Reused after Pushkaralu.** The same nodes serve Maha Shivaratri, Dasara, temple queues, bus stands and railway stations, and the approach transfers to other mass gatherings such as Krishna Pushkaralu.

Requirement (challenge §4, §10)	How the design meets it
High footfall, time-sensitive	Designed for 4+ persons/m^2; 1-second decision loop; warns minutes before the limit.
Usable by field personnel	Three colours, three actions (let through, slow or divert, hold). Police need no phone or app.
Multilingual access	Colour signals need no language; dashboard and staff messages to be provided in Telugu and English for the pilot.
Reliability on site	Decisions stay local; tested with 30–50% radio message loss, network cuts and dead boards.
Privacy and cybersecurity	Counts only: cameras count on the device; no facial recognition, no stored video, no personal data. Planned for the pilot: authenticated node messages and role-based access to the API.
Interoperability	Takes density from any source, including existing CCTV/AI counts. Documented REST API to the RMC ICCC.
Scale	Tested on a route graph of 68 compartments, 20 junctions and 10 Nagaras (244 boards). Adding a ghat means adding nodes and edges to the route map.
Post-event use	Same nodes for Maha Shivaratri, Dasara and other festivals, temple queues, bus-stand and railway-station crowds.

9 · IMPLEMENTATION PLAN (ALIGNED WITH CHALLENGE §11) · DETAILS IN THE COMPANION BUDGET AND PLAN

When	What	Measured
Nov 2026 – Jan 2027	Map routes for the Kotilingala model compartment; install counters and nodes; connect CCTV counts and the ICCC	Counting accuracy vs manual counts
Feb – Mar 2027	**Live pilot at Maha Shivaratri**	Warning lead time, false alerts per day, radio delivery rate, node uptime
Mar – Jun 2027	Scale to 68 compartments, 20 junctions, 10 Nagaras; police drills	Drill response times
Jun – Jul 2027	12-day operation from the ICCC	Live KPIs, incident log

10 · INDICATIVE COST ESTIMATE · COST-EFFECTIVENESS

One gate or exit (counter) node	₹ (retail)
ESP32 board with LoRa radio (SX1276) [11]	1,650
Two time-of-flight counting sensors (VL53L1X) [12], shaded	1,130 – 1,330
Three-colour IP65 signal tower with buzzer [13]	630 – 3,135
IP65 enclosure [14], pole clamp, sun hood	410 – 800
Power: 12 V 7 Ah battery (about 15 h backup), adapter, wiring	1,300 – 2,000
Per counter node	**≈ 5,100 – 8,900**

Pilot (Kotilingala model compartment: 11 nodes and a gateway, with our own cameras): about **₹3.4–5.2 lakh** all-in, including GST, installation, operations and validation.

Full site (244 nodes and 10 zone gateways, as in our site model): about **₹25–43 lakh** all-in if wide-area density comes from the CCTV the state is installing, or **₹41–64 lakh** with our own camera on each of the 108 wide-area nodes. That is under 0.02% of the ₹4,026 crore of Pushkaralu works [16].

Software is open-source and runs on one server or VM at the ICCC.

Indicative only: unit prices are public retail listings (November 2026); the separate budget plan gives every line, its source and its assumption. Full-site spending follows a go / no-go review of the pilot. Nodes are reused after the event.

11 · LIMITATIONS AND RISKS

- **All results so far are from simulation.** The compartment layout is assumed until we receive the Kotilingala model-compartment plans, and the sensor errors are modelled, not yet measured; the pilot measures them.
- **Wide roads depend on a density reading** (CCTV or edge camera): at least every 2 minutes with ≤ 20% error. A camera-free radio fallback (signal attenuation across the road) was safe only when it reported every 10 s, with no margin.
- A command centre with a working network is slightly better on a surge (peak 2.9 vs 3.3 persons/m^2); that is why we propose the hybrid.
- Two neighbouring boards failing together (gate and junction) can leave a gap in 2 of 5 tests; the ICCC covers that case when the network is up.
- In about 1 in 4 near-miss scenarios, where the crowd rose fast but stopped short of danger, the system slowed the flow as a precaution.
- Line counters work only in lanes, so they are placed only where barricades already form them (gates, exits); the VL53L1X needs shade, since direct sun cuts its range to under 1 m. Nodes need a power, weather, theft and maintenance plan, which is part of the pilot design.

12 · SUPPORT REQUIRED FROM THE ORGANISERS AND GOVERNMENT STAKEHOLDERS

1. Kotilingala model-compartment layout, planned pedestrian routes and ghat capacities, to build the route map.
2. Access to existing CCTV crowd-count feeds and the ICCC/RTGS integration interface.
3. Permission and police coordination for a live pilot at Maha Shivaratri (Feb–Mar 2027), including mounting points and power at gates and junctions.
4. Police and volunteer time for drills on the three-colour protocol.
5. A server or VM in the RMC or state data centre.
6. Procurement support for 244 nodes and 10 zone gateways.

13 · TEAM

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Also submitted: PS-03 lost-person reunification, which reuses the same nodes.

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